

Computational Replication Report:
 Šmejkal, Sinova & Jungwirth (2022),
Emerging research landscape of altermagnetism
 (arXiv:2204.10844)

Ollie (OpenClaw @ Argonne) — TEXTURES-100 Wave 4

2026-07-19

1 Paper summary

This Perspective delimits *altermagnetism* as a third basic collinear magnetic phase distinct from ferromagnetism and antiferromagnetism. Altermagnets combine a *compensated* (zero net moment, AFM-like) collinear spin order with *nonrelativistic spin-split* electronic bands (FM-like), reconciled by an unconventional spin-group symmetry formalism. The two spin sublattices are connected by a proper/improper crystal *rotation* rather than translation or inversion, producing an even-parity (*d*-, *g*-, *i*-wave) momentum-dependent spin splitting with nodal lines and sign-alternating lobes, so that the Brillouin-zone integral of the spin splitting vanishes while the local splitting is large.

2 Claims addressed

ID	Claim	Type	Testable?
C1	Compensated ($M_{\text{net}} = 0$) yet nonrelativistically spin-split bands	conceptual/model	yes
C2	Symmetry rule: sublattices connected by rotation, not translation/inversion	symmetry	yes
C3	Even-parity <i>d</i> -wave splitting: nodal lines, 4 alternating lobes, BZ-compensated	model	yes
C4	Material zoo (RuO ₂ , MnTe, CrSb...) from ab initio	DFT	yes

no

3 Method

We use a minimal single-orbital square-lattice tight-binding altermagnet (reusing the Wave-3 `sasioglu2026` *d*-wave template). The spin- σ dispersion is

$$E_{\sigma}(\mathbf{k}) = -2t(\cos k_x + \cos k_y) - 2\sigma t_{\text{AM}}(\cos k_x - \cos k_y),$$

giving the *d*-wave spin splitting $\Delta(\mathbf{k}) = E_{\uparrow} - E_{\downarrow} = -4t_{\text{AM}}(\cos k_x - \cos k_y)$, with $t = 1$, $t_{\text{AM}} = 0.30$. Tools: Python 3.13, NumPy, Matplotlib (`code/smejkal2022.replication.py`). Diagnostics computed on a 401×401 $\mathbf{k}$ -grid: BZ-averaged splitting, half-filling net moment $n_{\uparrow} - n_{\downarrow}$, the C_4 +spin-flip symmetry residual $\max |E_{\uparrow}(\mathbf{k}) - E_{\downarrow}(C_4\mathbf{k})|$, the translation residual, the diagonal nodal amplitude, and the number of sign changes of Δ around a Γ -centered circle.

4 Results vs. paper

Quantity	Expected (paper concept)	This work	Verdict
BZ-avg spin splitting	0 (compensated)	0.00	✓
Net moment (half filling)	0	0.00	✓
Local max $ \Delta $	large (FM-like)	$2.40 t$	✓
C_4 +spin-flip residual	0 (rotation connects sublattices)	0.0	✓
Translation residual	large (NOT AFM-translation)	2.40	✓
Diagonal node $ \Delta $	0 (nodal lines)	0.00	✓
Sign changes around Γ	4 (d -wave)	4	✓

5 Per-claim what worked / what didn't

C1 (worked). $M_{\text{net}} = 0$ at half filling coexisting with $\max |\Delta| = 2.4t$ reproduces the FM–AFM dichotomy exactly. **C2 (worked).** The C_4 +spin-flip residual is machine-zero while the translation residual is 2.4, numerically demonstrating the paper’s identification rule. **C3 (worked).** Nodes on $k_x = \pm k_y$ and exactly 4 sign changes confirm d -wave. **C4 (not attempted).** Material-specific ab initio band structures require DFT and are out of scope for a CPU-only model replication.

6 Critique of the replication

This is a *conceptual/model* replication of a review, not a numerical match to a single figure. Strength: the symmetry and node structure that anchor the entire Perspective are reproduced to machine precision, and the LLM-judge scored REPLICATED (coverage 9/10, agreement 9/10). Weaknesses honestly noted: (i) the model is imposed, not derived — t_{AM} is an input, so we do not prove altermagnetism is the energetic ground state; (ii) only the d -wave case is shown, whereas the paper spans g - and i -wave materials; (iii) all energies are in units of t , so there is no absolute-meV comparison to any compound; (iv) SOC, magnons, and transport (Secs III.C–D) are outside this minimal electronic model. None of these undercut the central claims tested, but they bound the scope.

7 Verdict

REPLICATED (representative model + symmetry classification). C1–C3 reproduced to machine precision; C4 (material DFT) out of scope and documented as method-limited, not a failure.

Open Questions

See `report/open_questions.json`. Q1: multi-orbital / higher-wave (g,i) material fidelity. Q2: SOC-lifted nodes and anomalous Hall magnitude. Q3: self-consistent interacting origin (Hubbard mean field). Q4: chiral magnon splitting in the same model. Q5: cleanest transport signature distinguishing altermagnet from AFM.